

Aliah University

Odd-Semester (Autumn) Examination - 2022

(For 3rd Semester BTech, CSE)

Paper Name: Discrete Mathematics

Paper Code: MATUGBS04

Full Marks: 80

Time: 3 Hrs

Group - A

(10X1=10)

1. Mention whether statements are True/False –

- All cyclic graphs are Hamiltonian.
- Every Bi-partite graph forms an odd length cycle.
- A relation with only diagonal elements is an equivalent relation.
- Every map in this world could be colored by a minimum of 4 distinct colors and following the coloring properties.
- In a complemented lattice, no vertex can have more than one complement.
- A statement is a valid statement when it's always a tautology.
- In solving recurrence relation, number of roots we get $>$ order of that equation.
- To-sets are distributed lattices.
- A trail can have a self-loop in it.
- Composition of two relations follows commutative property.

Group - B

(Answer any 6 questions) (6X5=30)

- Write a short note on – Euler graph.
- Define “vertex cover”. Find the cardinality of the minimum vertex cover for Wheel graph of order n (W_n). Mention the main difference between maximal matching and perfect matching.
- Prove that – a graph with no odd length cycle is planar iff it follows the equation $e \leq 2n - 4$, where e = size of the graph, n = order of the graph.
- Define relation between two sets. Mention its various types with suitable examples.
- Define join and meet points in a Hasse diagram. What is a lattice? When a lattice is called a “Boolean Lattice”? Explain with examples.
- Consider $A = \{2, 4, 6, 9\}$, and $B = \{1, 2, 3, 4, 5, 6\}$, then determine the values of $A \cup B$, $A - B$, $B - A$ and power set of A .
- In a university there are 2 types of students. Type1 students always tell the truth and Type2 students always lie. A fair coin is given to a random student and is requested to do a toss and asked to keep the result hidden till he/she was asked to tell. Upon asking, he/she replies - “The result of the toss is head if and only if I am telling the truth”. Then conclude whether the result is head/tail.

Group - C

(Answer any 4 questions) (10X4=40)

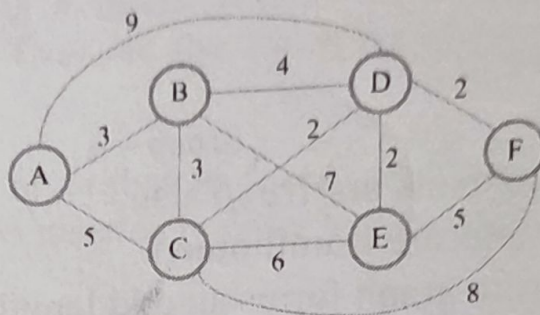
9.

- Define function and its types. Find total number of functions possible from Set A to set B , where $|A|=m$ and $|B|=n$.
- Consider a set $A = \{1, 2, 3\}$. Calculate number of symmetric relations on set A . A relation on set A is given as $R = \{(1, 2), (2, 1), (1, 1)\}$. Find the Reflexive and Transitive closure of R .

[5+5=10]

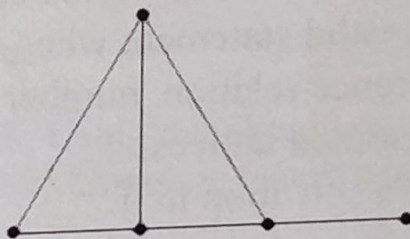
10.

- a) Prove – “In a non-trivial simple graph, there are at-least 2 vertices of the same degree”.
- b) A connected weighted graph is given below. Find a shortest path between A and F [4+6=10]
by using Dijkstra's Algorithm.



11.

- a) Find the chromatic polynomial for the following graph. If 5 colors are available, in how many ways can the vertices of this graph be properly colored?



- b) What is a Binary Search Tree? Insert values given below (in the order) and form the Binary Search Tree (BST) - Values are (in order): 42, 28, 17, 23, 13, 85, 36, 19, 81 [4+6=10]
Perform deletion of 19 and 42 and then show final BST.

12.

- a) Define tautology, contradiction and contingency. Give examples. Mention whether the expression: $((p \rightarrow q) \wedge (p \rightarrow r)) \rightarrow (p \rightarrow (q \wedge r))$ is a tautology or not.
- b) Explain quantifiers with examples. Suppose there are predicates - Graph(x): x is a graph, Connected(x): x is connected. Write a first order logic statement for – “Not every graph is connected.” [5+5=10]

13.

- a) Solve the recurrence relation: $a_n = 4a_{n-1} + 5a_{n-2}$, $a_1 = 2$ and $a_2 = 6$ [5+5=10]
- b) Consider the set $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and the relation divides (/) to make a Poset $P = (D_{30}, /)$. Draw the Hasse Diagram for P. Is it a lattice? If yes, then find complements of 3 and 15.